

ANIK BARURY

CS2604.

Q1: Let V be a vector space and let v_1, v_2, v_3 be linearly independent.

For $a \in \mathbb{F}$,

$$\text{define } w_1 = v_1 + v_2 \quad w_2 = v_2 + v_3 \quad w_3 = v_3 + av_1.$$

Determine all values of a for which w_1, w_2, w_3 form a basis of $\text{span}(v_1, v_2, v_3)$.

For w_1, w_2, w_3 to be span of v_1, v_2, v_3 , they have to be linearly independent.

So, for constant $c_1, c_2, c_3 \in \mathbb{F}$.

$$\Rightarrow c_1 w_1 + c_2 w_2 + c_3 w_3 = 0$$

$$\Rightarrow c_1(v_1 + v_2) + c_2(v_2 + v_3) + c_3(v_3 + av_1) = 0$$

$$\Rightarrow v_1(c_1 + ac_3) + v_2(c_1 + c_2) + v_3(c_2 + c_3) = 0$$

$$\left. \begin{array}{l} \Rightarrow c_1 + ac_3 = 0 \\ \Rightarrow c_1 + c_2 = 0 \\ \Rightarrow c_2 + c_3 = 0 \end{array} \right\} \begin{array}{l} \Rightarrow c_3(1+a) = 0 \quad \text{--- (i)} \\ \Rightarrow c_1 = -c_2 \\ \Rightarrow c_2 = -c_3 \end{array} \quad \left. \begin{array}{l} \vdots v_1, v_2, v_3 \neq 0 \\ \text{and} \end{array} \right\}$$

$$\text{for (eq. i)} \quad c_3(1+a) = 0$$

we have two case i) $c_3 = 0$ OR ii) $(1+a) = 0$

for eq. i to be 0. But $c_3 = 0$ is ~~not~~ always true as v_1, v_2, v_3 are LI

But if $(1+a) = 0$, then for non zero c_3 , $c_3(1+a) = 0$ is the vector v_1, v_2, v_3 are not LI.

So for them to be a basis, we must ensure, $a+1 \neq 0 \Rightarrow \boxed{a \neq -1}$.

So for v_1, v_2, v_3 be a basis we must ensure $a \neq -1$.

b. For exceptional values of a , determine $\dim(\text{span}(w_1, w_2, w_3))$.

For exceptional values of a , ie $a = -1$

$$\text{we have, } w_1 = v_1 + v_2$$

$$w_2 = v_2 + v_3$$

$$w_3 = v_3 + (-1)v_1.$$

If we do, $w_1 - w_2 + w_3$ is

$$\Rightarrow v_1 + v_2 - v_2 - v_3 + v_3 - v_1 = 0.$$

ie w_1, w_2, w_3 are linear dependent, so $\dim(\text{span}(w_1, w_2, w_3)) \neq 3$.

lets check for w_1, w_2 .

$$c_1 w_1 + c_2 w_2 = 0$$

$$\Rightarrow c_1(v_1 + v_2) + c_2(v_3 + v_2) = 0$$

$$\Rightarrow v_1(c_1) + v_2(c_1 + c_2) + v_3(c_2) = 0$$

$\Rightarrow c_1 = 0, c_2 = 0$. So w_1 and w_2 are linearly independent.

$$\text{So } \dim(\text{span}(w_1, w_2)) = \dim(\text{span}(w_1, w_2, w_3)) = 2.$$

2. Let $V = \mathcal{P}_4(\mathbb{R})$. and consider the following:-

$$p_1(x) = 1+x, \quad p_2(x) = x+x^2, \quad p_3(x) = x^2+x^3,$$

$$p_4(x) = x^3+x^4, \quad p_5(x) = 1+x^4.$$

a) Determine whether the five polynomials form a basis of $\mathcal{P}_4(\mathbb{R})$.

We know, for basis, we need linearly independent set of vectors.

considering $c_1, c_2, c_3, c_4, c_5 \in \mathbb{R}$.

$$\Rightarrow c_1 p_1(x) + c_2 p_2(x) + c_3 p_3(x) + c_4 p_4(x) + c_5 p_5(x) = 0.$$

$$\Rightarrow c_1(1+x) + c_2(x+x^2) + c_3(x^2+x^3) + c_4(x^3+x^4) + c_5(1+x^4) = 0$$

$$\Rightarrow (c_1+c_5) + x(c_1+c_2) + x^2(c_2+c_3) + x^3(c_3+c_4) + x^4(c_4+c_5) = 0.$$

$$\Rightarrow c_1 + c_5 = 0 \Rightarrow c_1 = -c_5.$$

$$c_1 + c_2 = 0 \Rightarrow c_1 = -c_2 \quad \left. \vphantom{\begin{matrix} c_1 + c_5 = 0 \\ c_1 + c_2 = 0 \end{matrix}} \right\} c_5 = c_2 = c_1$$

$$c_3 + c_4 = 0 \Rightarrow c_3 = -c_4 \quad \left. \vphantom{\begin{matrix} c_3 + c_4 = 0 \\ c_2 + c_3 = 0 \end{matrix}} \right\} c_2 = c_4 = c_5$$

$$c_2 + c_3 = 0 \Rightarrow c_3 = -c_2$$

$$c_4 + c_5 = 0 \Rightarrow 2c_4 = 0 \Rightarrow c_4 = 0 \quad \text{ie } \boxed{c_1 = c_2 = c_3 = c_4 = c_5 = 0}$$

So it's a trivial case, i.e. the vectors form a basis for $\mathcal{P}_4(\mathbb{R})$.

b) If they do not, find a maximal independent subsets.

Yes, they are a basis.

c) Show that the vectors we discard can nevertheless be recovered from the vectors you are kept.

Are linearly independent, i.e. we can't remove anything without reducing the dimension.

d) Give a basis of $P_4(\mathbb{R})$ consisting of linear combinations of $p_1, \dots, p_5$ but not necessarily consisting of members of the original list.

Any scaling of our basis vectors remains linearly dependent.

$$p_1'(x) = 2 + 2x; \quad p_2'(x) = 2x + 2x^2; \quad p_3'(x) = 3x^2 + 3x^3$$

$$p_4'(x) = 2x^3 + 2x^4; \quad p_5'(x) = \frac{1}{2} + \frac{x^4}{2}.$$

So, $\{p_1', \dots, p_5'\} = B'$ is our new Basis.

3. Let $B = \{v_1, v_2, v_3, v_4\}$ be a basis of a vector space V .

with defined, $u_1 = v_1 + v_2$, $u_2 = v_2 + v_3$, $u_3 = v_3 + v_4$.

a) Prove that u_1, u_2, u_3 are linearly independent.

Consider $c_1, c_2, c_3 \in \mathbb{F}$.

$$\text{we know, } c_1 u_1 + c_2 u_2 + c_3 u_3 = 0$$

$$\Rightarrow c_1 v_1 + c_2 (v_1 + v_2) + c_3 (v_2 + v_3) + c_4 (v_3 + v_4) = 0.$$

$$\Rightarrow \boxed{c_1 = 0} \Rightarrow c_2 + c_3 = 0 \Rightarrow c_2 = -c_3.$$

$$\Rightarrow c_1 + c_2 = 0 \Rightarrow \boxed{c_3 = 0} \Rightarrow \boxed{c_2 = 0}$$

$$\Rightarrow c_1 = -c_2$$

So, u_1, u_2, u_3 are linear independent.

b) Find all vectors $u_4 \in V$ for which u_1, u_2, u_3, u_4 is a basis of V .

Let our new vector u_4 be $a\bar{v}_1 + b\bar{v}_2 + c\bar{v}_3 + d\bar{v}_4$.

considering $c_1, c_2, c_3, c_4 \in \mathbb{F}$.

$$\Rightarrow c_1 u_1 + c_2 u_2 + c_3 u_3 + c_4 u_4 = 0.$$

$$\Rightarrow c_1 + a c_4 = 0 \Rightarrow c_1 = -a c_4.$$

$$\Rightarrow c_1 + c_2 + b c_4 = 0 \Rightarrow c_2 = (a-b) c_4.$$

$$\Rightarrow c_2 + c_3 + c c_4 = 0 \Rightarrow c_3 = (-a+b-c) c_4.$$

$$\Rightarrow c_3 + d c_4 = 0 \Rightarrow c_4 \times (-a+b-c+d) = 0.$$

Same logic, we need $(-a+b-c+d) \neq 0$ as if it equals to 0 for a non zero value of c_4 we will get our solutions.

$$\text{So, } -a+b-c+d \neq 0$$

$$\Rightarrow \boxed{b+d \neq a+c} \text{ for all } a, b, c, d \text{ we will get } u_4 \text{ such that}$$

u_1, u_2, u_3, u_4 forms a basis.

c) Among ~~four~~ possible choices, determine which u_4 lie in $\text{span}(v_1, v_2, v_3)$

For u_4 to lie in $\text{span}(v_1, v_2, v_3)$, we can write

$$u_4 = a v_1 + b v_2 + c v_3.$$

Comparing the above eqn with part (b), we notice,

our $d = 0$, as we don't have any v_4 part.

$$\Rightarrow -a+b-c+(0) \neq 0$$

$$\Rightarrow \boxed{b \neq a+c} \forall a, b, c \text{ we can guarantee } u_4 \in \text{span}(v_1, v_2, v_3).$$

d) Explain why part (b) is an useful application rather than mere computation.

In given problem, we have a vector space V where we atleast have 4 LI vectors (v_1, v_2, v_3, v_4) ie the $\dim(V) = 4$.

In part (a) we made a new set of LI vectors from L. Comb. of $v_1, \dots, v_4$ as u_1, u_2, u_3 , so $\text{span}(u_1, u_2, u_3) = V'$ (let) where $V' \subsetneq V$.
and $\dim(V') = 3$.

We found a new Linearly independent vector u_4 from v_1, v_2, v_3, v_4 and we add it in set u_1, u_2, u_3 such that $\text{span}(u_1, u_2, u_3, u_4) = V''$ (let)
and $\dim(V'') = 4$.

So we ~~ex~~ 'EXTEND' our given BASIS' to a higher order/dim space.

We can conclude this for any given smaller set to form of a larger or full basis.

Q4. Suppose U, W are subspaces of a finite dim vector space V and $\dim(U) = 7$, $\dim(W) = 6$, $\dim(V) = 10$.

a) Smallest possible value of $\dim(U+W)$?

When we add $(U+W)$ our dim will be atleast the largest one out of U, W . ie 7.

Largest possible value of $\dim(U+W)$?

As U and W are subspaces of V we can't get more than $\dim(V)$ ie 10.

a) Smallest value of $\dim(U \cap W)$.

we know,

$$\dim(U+W) = \dim(U) + \dim(W) - \dim(U \cap W).$$

$$\text{So, } \dim(U \cap W) = \dim(U) + \dim(W) - \dim(U+W).$$

$$= 7+6 - (10) = \boxed{3}$$

largest value of $\dim(U \cap W)$.

$$\text{again, } \dim(U \cap W) = \dim(U) + \dim(W) - \dim(U+W)$$

$$= 7+6 - 7 = \boxed{6}$$

b) For extreme values, can these $U \cap W$ exist?

Yes, for max intersection i.e 6, we can imagine as $W \subset U$ or $U \subset W$. Then $W \cap U = W$.

for min intersection, i.e 3, we can imagine as for V we have 10 LI vectors, build U with first 7, W with remaining 3 and then (3 common) from U .

d) If we define $U+W = V$, find $\dim(U \cap W)$?

CASE 1.

If we give full ^{unique} rank to vectors to U from V i.e U has 7 unique vectors. To generate/cover the whole space V we need to have the remaining 3 unique vectors from V to W and as $\dim(W)=6$, we need to have common 3 vectors from U .

$$\text{i.e } U \cap W = 3.$$

CASE II

Similarly, if we give full unique set to W is 6, U must have 1 remaining vectors and as $\dim(U) = 7$ we again need to have 3 common vectors, i.e. $U \cap W = 3$.

So in both of the cases $\boxed{U \cap W = 3}$.

5) Let U, W, Z be subspaces of V , and suppose $V = U \oplus W$ and $W \subseteq Z \subseteq U$. Again $U \cap Z = \{0\}$.

a) Proof or disprove : $V = U \oplus Z$.

We need to show: $\Rightarrow V = U + Z$ $\Rightarrow U \cap Z = \{0\}$.

The second condition is already given, $U \cap Z = \{0\}$.

We have, $V = U \oplus W$ i.e. $V = U + W$

Let's say $\bar{v} \in V$, $\bar{u} \in U$, $\bar{w} \in W$.

$$\text{So, } \bar{v} = \bar{u} + \bar{w}.$$

Also, $W \subseteq Z$, so $\bar{w} \in Z$.

$$\Rightarrow \bar{v} = \bar{u} + \bar{w} \subseteq U + Z.$$

$\Rightarrow V \subseteq U + Z$, But as U and Z are subspaces of V itself.

We can't get more than V .

$$\Rightarrow \boxed{V = U + Z} \text{ hence, } V = U \oplus Z. \checkmark$$

b) Additional conditions for Z ?

We saw above that $U \cap Z = \{0\}$ mathematically forces the condition of $Z = W$. [$\because$ Z can't be diff bigger / contain unique vector other than v].

So, $Z = W \Rightarrow V = U \oplus Z$ to be true as $V = U \oplus W$ is given.

6. Let $T: V \rightarrow V$ be linear, and let $U \subseteq V$ be a subspace satisfying $T(\bar{u}) = \bar{u} \quad \forall \bar{u} \in U$. Also $T(\bar{v}) = 0 \quad \forall \bar{v} \in W$. and $V = U \oplus W$.

a) Determine $T(u+w)$ for arbitrary $\bar{u} \in U, \bar{w} \in W$.

We know, from the properties of linear transformation,

$$T(c\bar{a} + \bar{b}) = cT(\bar{a}) + T(\bar{b}).$$

$$\Rightarrow T(\bar{u} + \bar{w}) = T(\bar{u}) + T(\bar{w}) = \bar{u} + 0 = \bar{u} \in U.$$

b) Determine null T .

null of a linear transformation means set of all vector from input space (here V) which get maps to $\bar{0}$.

Here as $T(\bar{v}) = 0 \quad \forall \bar{v} \in W$, null $T = W$.

c) Determine Range T .

Range (T) means all vector set which retain its dimensions.

Here $T(\bar{u}) = \bar{u}, \bar{u} \in U$, Range (T) = U .

d) Determine whether T is invertible.

We know, every linear map is a function, and a function is invertible if it is bijective, i.e. both one-one and onto.

But for it to be one-one we must show null $T = \{0\}$.

But in our case null $T = W$.

So until $W = \{0\}$, T is not invertible.

e. Let Transformation T be,

$$T(\bar{u}) = 2\bar{u} \quad \bar{u} \in U \quad \text{and} \quad T(\bar{w}) = -\bar{w} \quad \bar{w} \in W$$

Determine the null space and range.

Null space: Let $v = u + w$.

$$T(v) = T(u + w) = T(u) + T(w)$$

$$\Rightarrow 2u - w \quad \text{let this } T(v) = 0$$

$$\text{ie } 2u - w = 0$$

$$\Rightarrow 2u = w \quad [\because \bar{u}, \bar{w} \text{ are vectors, a twice of a vector equals another vector means } \bar{0}].$$

$$\text{So, } u = w = 0 \quad \text{so } v = 0 \quad \text{so } \text{Null } T = \{0\}.$$

Range space: So $T(u/2) = u$ and $T(-w) = w$.

we find any arbitrary vector $v = u + w$ by $T(u/2 - w) = u + w$.

So $\text{Range } T = \text{whole space ie } V$.

f. Prove T^{-1} is feasible OR T is invertible.

M1 Given, $T(\bar{u}) = 2\bar{u}$ and $T(\bar{w}) = -\bar{w}$

$$\Rightarrow T^{-1}T(\bar{u}) = T^{-1}(2\bar{u}) \quad \Rightarrow T^{-1}T(\bar{w}) = T^{-1}(-\bar{w})$$

$$\Rightarrow \bar{u} = 2T^{-1}(\bar{u}) \quad \Rightarrow \bar{w} = -T^{-1}(\bar{w})$$

$$\Rightarrow T^{-1}(\bar{u}) = \bar{u}/2 \quad \Rightarrow T^{-1}(\bar{w}) = -\bar{w}$$

As, T^{-1} is existing, it is possible.

M2 T is invertible if it is a bijection, i.e.

• into : Yes as $\text{null } T = \{0\}$ (shown in (e)).

• onto : Yes as $\text{Range } T = V$ (shown in (e))

hence, T^{-1} is bi bijection, i.e. invertible.

7. Let $T: \mathcal{P}_3(\mathbb{R}) \rightarrow \mathcal{P}_3(\mathbb{R})$ defined by,

$$T(p) = p + p' \text{ where } p \in \mathcal{P}_3(\mathbb{R}) \text{ and } p' \text{ is first derivative of } p.$$

a) Prove that T is linear.

For linear, we need to show,

$$T(cp_1 + p_2) = cT(p_1) + T(p_2), \text{ where } p_1, p_2 \in \mathcal{P}_3(\mathbb{R}).$$

Let our Basis for $\mathcal{P}_3(\mathbb{R}) = \{1, x, x^2, x^3\}$ is standard basis.

$$\text{So, let } p_1 = a_1 + b_1x + c_1x^2 + d_1x^3.$$

$$p_2 = a_2 + b_2x + c_2x^2 + d_2x^3.$$

$$\text{LHS, } T(cp_1 + p_2) = T(ca_1 + cb_1x + cc_1x^2 + cd_1x^3 + a_2 + b_2x + c_2x^2 + d_2x^3).$$

$$= T(ca_1 + a_2 + (cb_1 + b_2)x + (cc_1 + c_2)x^2 + (cd_1 + d_2)x^3)$$

$$= ca_1 + a_2 + (cb_1 + b_2)x + (cc_1 + c_2)x^2 + (cd_1 + d_2)x^3 + (cb_1 + b_2)x + 2(cc_1 + c_2)x + 3(cd_1 + d_2)x^2.$$

$$= (ca_1 + cb_1 + a_2 + b_2) + (cb_1 + 2cc_1 + b_2 + 2c_2)x + (cc_1 + 3cd_1 + c_2 + 3d_2)x^2 + (cd_1 + d_2)x^3. \quad \text{--- (i)}$$

$$\text{RHS, } cT(p_1) + T(p_2).$$

$$= cT(a_1 + b_1x + c_1x^2 + d_1x^3) + T(a_2 + b_2x + c_2x^2 + d_2x^3).$$

$$= c\{a_1 + b_1x + c_1x^2 + d_1x^3 + b_1 + 2c_1x + 3d_1x^2\} + (a_2 + b_2x + c_2x^2 + d_2x^3 + b_2 + 2c_2x + 3d_2x^2)$$

$$= ca_1 + cb_1 + a_2 + b_2 + (cb_1 + 2cc_1 + b_2 + 2c_2)x + (cc_1 + 3cd_1 + c_2 + 3d_2)x^2 + (cd_1 + d_2)x^3 \quad \text{--- (ii)}$$

As we have $\text{LHS} = \text{RHS}$, we conclude it as linear.

b) Determine whether the subspace $U = P_1(\mathbb{R})$ is mapped into itself by T .

We know, into condition is a transformation $T: V_1 \rightarrow V_2$ when a vector $\vec{v} \in V_1$ and $T(\vec{v}) \in V_2$.

Checking $T: U \rightarrow U$.

considering a basis $B = \{1, x\}$ for $P_1(\mathbb{R})$

a vector $\vec{u} = a + bx \in U$.

$$T(\vec{u}) = a + bx + b = a + b + bx \Rightarrow c + bx \text{ if } c = a + b$$

is in the form of B . So $T(\vec{u}) \in U$ - ie into.

c) Determine whether U is mapped onto itself.

Onto means any $\vec{v}$ vector has a representation let $\vec{u} \mid T(\vec{u}) = \vec{v}$
 $\vec{v}, \vec{u} \in U$.

let our $\vec{v} = c + dx$. and let our initial vector $= \vec{u} = a + bx$.

$$\text{So, } T(\vec{u}) = \vec{v}$$

$$\Rightarrow T(a + bx) = a + b + bx = c + dx$$

$$= a + b + bx = c + dx.$$

$$\Rightarrow a + b = c \Rightarrow a = c - b = c - d.$$

$$b = d$$

$$\therefore \vec{u} = (c - d) + dx \xrightarrow{T} c + dx.$$

So T is onto also.

d) Determine the null T .

we know $T: \mathcal{P}_3(\mathbb{R}) \rightarrow \mathcal{P}_3(\mathbb{R})$

Basis for $\mathcal{P}_3(\mathbb{R})$ is $\{1, x, x^2, x^3\}$.

For null space we know, $T(\bar{x}) = 0 \quad \forall x \in V$.

Let $\bar{x} = a + bx + cx^2 + dx^3$.

$$T(\bar{x}) = a + bx + cx^2 + dx^3 + b + 2cx + 3dx^2 = 0$$

$$\Rightarrow a + b + x(b + 2c) + x^2(c + 3d) + x^3(d) = 0$$

$$\Rightarrow a + b = 0 \quad \Rightarrow a = 0$$

$$\Rightarrow b + 2c = 0 \quad \Rightarrow b = 0$$

$$\Rightarrow c + 3d = 0 \quad \Rightarrow c = 0 \quad \Rightarrow \bar{x} = \bar{0}$$

$$\Rightarrow d = 0$$

So $\exists$ only vector $\bar{0}$ which get reduce to $\bar{0}$

So $\text{Null } T = \{\bar{0}\}$.

e) Determine whether T is invertible or not.

As we proved $\text{Null } T = \{0\} \Rightarrow \dim(\text{Null } T) = 0$.

we know, $\dim(V) = \dim(\text{Null } T) + \dim(\text{Range } T)$.

$$\Rightarrow 4 = 0 + \dim(\text{Range } T) = 4.$$

So, $\dim(\text{Null } T) = 0 \Rightarrow$ into $\left. \begin{array}{l} \dim(\text{Range } T) = 4 \text{ (full)} \Rightarrow \text{onto.} \end{array} \right\} \rightarrow \text{bijection} \rightarrow T \text{ is invertible.}$

f) If $q \in \mathcal{P}_3(\mathbb{R})$, determine whether there exist a unique $p \in \mathcal{P}_3(\mathbb{R})$ satisfying $p + p' = q$.

Let there be two polynomials p_1, p_2 where.

$$p_1 + p_1' = q \quad \text{and}$$

$$p_2 + p_2' = q.$$

$$\Rightarrow p_1 + p_1' = p_2 + p_2'$$

$$\Rightarrow (p_1 - p_2) + (p_1' - p_2') = 0$$

$$\text{Let } p_1 - p_2 = h \Rightarrow h' = p_1' - p_2'$$

$$\Rightarrow h + h' = 0$$

from part e we show, if $T(\bar{x}) = 0 \Rightarrow \bar{x} = 0$ as $\text{null } T = \{0\}$.

$$\therefore h = 0 \Rightarrow p_1 = p_2$$

So $\exists$ only a unique p for which $T(p) = q$.

8. Let $T: V \rightarrow V$ be a linear operator on an n -dim vector space and $\dim(\text{null } T) = k$.

a) what is the largest possible ^{dim} sub-space $U \subseteq V$ on ~~for~~ which T is one to one?

$T: U \rightarrow U$, for T to be one to one, we know $\text{null } T = \{0\}$

So U can't include any dimension from $\text{null}(T)$.

$$\text{So } \dim(U) = n - k \text{ (max).}$$

b. Show that such a sub-space always exists.

c. Can U be chosen such that, $V = \text{null } T \oplus U$.

If we proved $U \oplus \text{null } T = V$ where always exists we automatically show $\exists U$ where $\dim(U) = n - k$.

Yes, U can always be chosen such that $V = \text{null } T \oplus U$.

• Let $\dim(\text{null } T) = k \rightarrow$ Basis of the null space

$$B_1 = \{v_1, \dots, v_k\}.$$

• By basis extension theorem, any finite linearly independent list can be extended to a basis of a finite dimensional vector space, by adding more linearly independent vectors.

Let Basis $B_2 = \{v_1, \dots, v_k, u_1, \dots, u_{n-k}\}$ for V .

• Let us define $U = \text{span}(u_1, \dots, u_{n-k})$.

• As the set B_2 is basis for V , the combination of two span generates V as $V = \text{span}(v_1, \dots, v_k) + \text{span}(u_1, \dots, u_{n-k})$.

$$\Rightarrow V = \text{null } T + U. \quad - (i)$$

• And because it is a basis, the vectors are strictly linearly independent in $\text{null } T \cap U = \{0\}$ - (ii)

from eq. i, ii we conclude $V = \text{null } T \oplus U = V$.

d) If $k > 0$, can $T: U \rightarrow V$ ever be onto?

Given, $k > 0$. we know $\max \dim(U) = n - k$.

we know T is a function from U to V .

$$U \xrightarrow{T} V$$

where $\dim(U) = n - k$ and $\dim(V) = n$.

As $k > 0 \Rightarrow n - k < n$.

But for onto, from $A \rightarrow B$ we need $|A| \geq |B|$.

Here we don't have $\dim(U) \geq \dim(V)$.

So $T: U \rightarrow V$ is never onto if $k > 0$.

e) If $k = 0$, what structural conclusion can you draw about T .

If $k = 0 \Rightarrow \dim(\text{null } T) = 0$ i.e. $\text{null } T = \{0\}$.

we know, if a linear map T has $\text{null } T = \{0\}$.

it means, T is into. (one to one).

And if a linear map T maps from finite dim to finite dim vector space and T is into it means T is onto.

And if T is into and onto $\rightarrow$ bijection or invertible.

$\therefore T: U \rightarrow V$ is invertible $\Rightarrow$

$$\dim(U) = \dim(V)$$

$$n - k = n \Rightarrow \boxed{k = 0} \text{ Given condition.}$$

9. Let U, W be a finite-dimensional sub-spaces of V .
 Suppose $\dim(U) = r$, $\dim(W) = s$, $\dim(U \cap W) = k$.

a) Prove that there exists a basis of V of the form.

$$u_1, \dots, u_k, u_{k+1}, \dots, u_r, w_{k+1}, \dots, w_s, z_1, \dots, z_m.$$

where

$u_1, \dots, u_k$ is a basis of $U \cap W$,

$u_1, \dots, u_r$ is a basis of U and

$u_1, \dots, u_k, w_{k+1}, \dots, w_s$ is a basis of W .

Proof:-

• Let $U \cap W$ be a subspace of V . Let $\{u_1, \dots, u_k\}$ be a basis for $U \cap W$.

• Extending to U , because $U \cap W \subseteq U$, the set $\{u_1, \dots, u_k\}$ is linearly independent. By the basis extension theorem, we can append $(r-k)$ vectors to form basis of U : $\{u_1, \dots, u_k, u_{k+1}, \dots, u_r\}$.

• Extending to W , similarly $U \cap W \subseteq W$, we can extend our set $\{u_1, \dots, u_k\}$ to form basis of W by, $\{u_1, \dots, u_k, w_{k+1}, \dots, w_s\}$.

• Is our new set, i.e. mixture of vectors from U, W linearly independent?

Assuming a linear combination over field F :

$$c_1 \bar{u}_1 + \dots + c_r \bar{u}_r + d_{k+1} \bar{w}_{k+1} + \dots + d_s \bar{w}_s = 0.$$

$$\Rightarrow c_1 \bar{u}_1 + \dots + c_r \bar{u}_r = - (d_{k+1} \bar{w}_{k+1} + \dots + d_s \bar{w}_s)$$

Here our LHS vector (let $\bar{u}'$) is completely in U and RHS vector (let $\bar{v}'$) completely in W .

But $\bar{u}' = \bar{v}'$

So, $\bar{u}' / \bar{v}' \in \mathcal{U} \cap \mathcal{W}$.

Hence we can write $\bar{u}' / \bar{v}'$ solely as a combination of Basis($\mathcal{U} \cap \mathcal{W}$)

$$\Rightarrow -(d_{k+1}\bar{w}_{k+1} + \dots + d_s\bar{w}_s) = a_1\bar{u}_1 + \dots + a_k\bar{u}_k.$$

$$\Rightarrow a_1\bar{u}_1 + \dots + a_k\bar{u}_k + d_{k+1}\bar{w}_{k+1} + \dots + d_s\bar{w}_s = 0.$$

So As, $\bar{u}_1, \dots, \bar{u}_k$ is a basis for $\mathcal{U} \cap \mathcal{W} \Rightarrow a_1 = \dots = a_k = 0$.

This forces $d_1 = \dots = d_s = 0$.

$$\text{Similarly, } (c_1\bar{u}_1 + \dots + c_r\bar{u}_r) = a_1\bar{u}_1 + \dots + a_k\bar{u}_k$$

$$\Rightarrow a_1 = \dots = a_k = 0 \Rightarrow c_1\bar{u}_1 + \dots + c_r\bar{u}_r = 0$$

As, $(u_1, \dots, u_r)$ is a basis for $\mathcal{U}$; $\Rightarrow c_1 = \dots = c_r = 0$

So, our set is linearly independent.

• Extending to $\mathcal{V}$, we made $\mathcal{U} \cup \mathcal{W}$ now we should add the remaining vectors (let m) to our set to completely our set of Basis for $\mathcal{V}$,

$$\text{Basis}(\mathcal{V}) = \{u_1, \dots, u_k, u_{k+1}, \dots, u_r, w_{k+1}, \dots, w_s, z_1, \dots, z_m\}.$$

b) Using above proof to prove $\therefore \dim(U+W) = r+s-k$.

By definition, the subspace $U+W$ is spanned by the union of any basis of U and any basis of W .

Using the bases defined in part (a), i.e. $S = \{u_1, \dots, u_r, w_{k+1}, \dots, w_s\}$

we also showed S is linearly independent.

Because S spans $U+W$ and is linear independent, S is a basis for $U+W$.

The dimension is simply the total no. of vectors in S .

$\rightarrow$ There are r vectors in the sequence $u_1, \dots, u_r$.

$\rightarrow$ There are $(s-k)$ vectors in the sequence $w_{k+1}, \dots, w_s$.

$$\therefore \dim(U+W) = r + (s-k) = r+s-k.$$

c) Explain exactly where the basis-extension theorem is being used.

In part (a), we have used basis extension theorem thrice

- Extending the linear independent set $\{u_1, \dots, u_k\}$ to form the basis for U .
- Extending that same set $\{u_1, \dots, u_k\}$ to form the basis for U .
- Extending the combined linearly independent set $\{u_1, \dots, u_r, u_{k+1}, \dots, w_s\}$ to form the final basis for V .

d. Deduce the criterion for $U+W = U \oplus W$.

By definition, the sum of two sub-space $U+W$ is a direct sum is $U \oplus W$ iff their intersection is trivial i.e. $U \cap W = \{0\}$.

If $U \cap W = \{0\}$, then $\dim(U \cap W) = 0$.

So, $K=0$.

Hence,
~~So~~ $\dim(U+W) = r+s \Rightarrow U \oplus W = V$.

10. Let V be a 6 dimensional vector space.

We want to construct a linear operator $T: V \rightarrow V$

satisfying all of the following:

- $\dim(\text{null } T) = 2$ - $\dim(\text{Range } T) = 4$ and $\dim(\text{null } T \cap \text{Range } T) = 1$.

a) Prove that such operator can exist.

By the rank-Nullity Theorem, a linear operator $T: V \rightarrow V$.

on finite dim. space $\dim(\text{null } T) + \dim(\text{Range } T) = \dim(V)$.

Our given data, $\dim(\text{null } T) + \dim(\text{Range } T) = \dim(V) \Rightarrow 2+4=6$.

Also $\dim(\text{null } T \cap \text{Range } T) \leq \min(\overset{\dim}{\text{null } T}, \dim \text{Range } T)$.

and we have $1 \leq \min(2, 4)$.

So, as there are not structural contradictions, such an operator can exist.

b) Construct one explicitly by choosing a basis $= \{v_1, v_2, v_3, v_4, v_5, v_6\}$.

Given, Basis of $T = \{v_1, v_2, v_3, v_4, v_5, v_6\}$.

$\begin{aligned} & \cdot T(v_1) = 0 \\ & \cdot T(v_2) = 0 \end{aligned} \left. \vphantom{\begin{aligned} & \cdot T(v_1) = 0 \\ & \cdot T(v_2) = 0 \end{aligned}} \right\} \dim(\text{null}(T)) = 2.$

$\cdot T(v_3) = v_1 \left. \vphantom{\cdot T(v_3) = v_1} \right\} \dim(\text{null } T \cap \text{Range } T) = 1$

$\cdot T(v_4) = v_4$

$\cdot T(v_5) = v_5$

$\cdot T(v_6) = v_6$

$\left. \vphantom{\begin{aligned} & \cdot T(v_4) = v_4 \\ & \cdot T(v_5) = v_5 \\ & \cdot T(v_6) = v_6 \end{aligned}} \right\} \dim(\text{Range } T) = 4.$

c) verify all three properties required.

From our construction in part (b).

1. $\dim(\text{Range } T) = 4.$

The range is the span of the outputs: $\text{span}(0, 0, v_1, v_4, v_5, v_6).$
is $\text{span}(v_1, v_4, v_5, v_6)$. As ~~the~~ these four vectors are part of our original basis, they are linearly independent.

2. $\dim(\text{null } T) = 2.$

If we set $T(c_1 v_1 + \dots + c_6 v_6) = 0$, we get $c_3 v_1 + c_4 v_4 + c_5 v_5 + c_6 v_6 = 0 \Rightarrow c_3 = c_4 = c_5 = c_6 = 0$ so $c_1 v_1$ and $c_2 v_2$ are left free to be anything. So the null space is exactly $\text{span}(v_1, v_2)$ is $\dim(\text{null } T) = 2.$

3. $\dim(\text{null } T \cap \text{Range } T) = 1.$

The null space is generated by $\{v_1, v_2\}$. The range is generated by $\{v_1, v_4, v_5, v_6\}$. Only common vector is v_1 .

So ~~dim~~ $\text{null } T \cap \text{Range } T = \text{span}\{v_1\}$ is dim of 1.

d) Construct a space with $\text{null } T \cap \text{Range } T = \{0\}.$

• we have $\dim(\text{null } T) = 2$, $\dim(\text{Range } T) = 4.$

$$\begin{array}{l} \hookrightarrow T(v_1) = 0 \\ \hookrightarrow T(v_2) = 0 \end{array} \left. \begin{array}{l} \text{null } T \\ \text{span}(v_1, v_2) \end{array} \right\} \begin{array}{l} \hookrightarrow T(v_3) = v_3 \\ \hookrightarrow T(v_4) = v_4 \\ \hookrightarrow T(v_5) = v_5 \\ \hookrightarrow T(v_6) = v_6 \end{array} \left. \begin{array}{l} \text{Range } T \\ \text{span}(v_3, v_4, v_5, v_6) \end{array} \right\}$$

e) Explain why the data do not determine relative position?

The data given is, $\dim(\text{null } T) = 2$ and $\dim(\text{Range } T) = 4$.

Only tells us about size of the sub-space.

As we shown in part (b) and (d) that any thing can be made in the intersection can be 1D/0D. ie the intersection of the spaces are difficult to determine solely on the size of sub-spaces.

f) Determine whether either operator can be invertible.

A linear transformation on a finite-dim space is invertible iff it is a bijection (into + onto).

- For an map to be into (one-to-one) the null space = $\{0\}$
- For a map to be onto (surjective) the $\text{Range } T = \text{dim } V$.

But $\dim(\text{null } T) = 2$ is not 0 and

$\dim(\text{Range } T) = 4$ is not 6 so

T will never be invertible.

4e) Suppose instead that $U \cap W = \{0\}$ what can we conclude

about whether $U + W = V$.

If $U \cap W = \{0\} \Rightarrow \dim(U+W) = \dim(U) + \dim(W) - \dim(U \cap W)$

$$\Rightarrow 10 \neq 6 + 7 + 0 = 13$$

So mathematically it is impossible to have $U \cap W = \{0\}$.